

Deva Anand

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Pathos
Machine Learning Engineer Intern Hiring Team
New York, NY

Dear Hiring Team,

I'm applying for the Machine Learning Engineer Intern role. I'm a Computer Science master's student at UMass Amherst, and I want to be upfront about where I fit and where I'd be learning, because I'd rather you know exactly what you're getting.

Where I fit cleanly is two of your four focus areas: **Knowledge Graph + RAG** and **evaluation of reasoning**. I led the multi-view retrieval component of a team knowledge-graph-guided RAG pipeline for multi-hop QA, which lifted supporting-fact recall by 2.68% over the KG²RAG baseline on a 4,905-question evaluation. And evaluation is something I keep coming back to: I built **AutoEval**, an agent that does dataset curation and metric design, proposing test cases, scoring systems on a discriminative-power objective, and accepting only updates that improve it, with guardrails so it can't game its own benchmark. In a separate interpretability study I designed careful metrics and reported a clean null with confidence intervals rather than forcing a positive result. Rigorous, honest evaluation is the part of ML I care about most, and it maps directly to your reasoning-evaluation work.

On fundamentals, I implemented a reverse-mode autodiff engine from scratch (validated against JAX) and trained and compared architectures and optimizers in PyTorch, so I understand training workflows from the ground up, not just as API calls. I also have real exposure to medical and multimodal data: my first publication applied transfer learning to neuroimaging for disease classification, and I spent two years building backend systems on clinical healthcare data, which is part of why an oncology foundation model aimed at real patient outcomes resonates with me.

Now the honest part: I have not done hyper-scale GPU training or inference systems, no NCCL weight broadcast, no Mixture-of-Experts parallelism on H200s, no KV-cache disaggregation. That's a real gap, and I'm not going to pretend otherwise. What I'd bring instead is someone who learns fast in ambiguous spaces, ships, and has the systems-and-evaluation instincts to ramp on that infrastructure quickly. I've profiled and tuned a 1.4-billion-row data pipeline for throughput and tail-latency, so the performance mindset is there even if the GPU-specific tooling isn't yet.

I'd be glad to contribute on the KG/RAG and evaluation side from day one while learning the training-systems side from your team. Thank you for considering my application.

Sincerely,
Deva Anand